

Day: 12

* IP Datagram Structure

- The IP header contains everything a router needs to forward a packet.

Field	Size	Purpose
Version	4 bits	IPv4 or IPv6
Header Length	4 bits	Size of header
TTL	8 bits	Hops remaining
Protocol	8 bits	TCP = 6, UDP = 17, ICMP = 1
Source IP	32 bits	Where packet came from
Destination IP	32 bits	Where packet is going
Checksum	16 bits	Error detection
Flags	3 bits	Fragmentation control
Fragment Offset	13 bits	Position in fragment

* TTL — Time to Live

- TTL is one of the most important fields for DFIR
- Every router that forwards a packet

decreases TTL by 1.

- When TTL reaches 0 → router sends ICMP TTL Exceeded back
- This is exactly how traceroute works!

* Common Starting TTL Values:

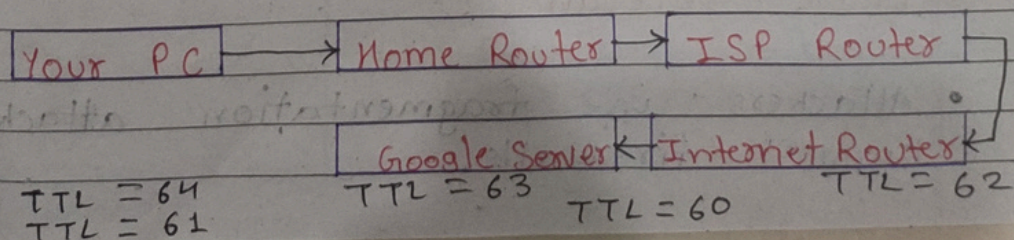
- Windows → 128
- Linux → 64
- Cisco / Network → 255

* DFI R Use:

- TTL tells us how far away the source is.
- Usually ~~the~~ Unusually low TTL = packet has traveled many hops.
- TTL = 1 packet flood = TTL Attack.

* How Routing Works

- A router's job is simple, → look at destination IP, find best path, forward.



* Routing Table

- Every router has one routing table.

Destination	Next Hop	Interface
192.168.1.0/24	Direct	eth0
0.0.0.0/0	192.168.1.1	eth0

- The 0.0.0.0/0 entry is the default route — "If you don't know where to send it, send it here".

* IP Fragmentation

- When a packet is too large for a network link, it gets fragmented (split into smaller pieces).
- MTU (Maximum Transmission Unit) = max packet size a link can carry.
- Ethernet MTU = 1500 bytes
- If packet > MTU → fragmented into smaller pieces.

* DFIR relevance:

- Attackers use fragmentation attacks to bypass firewalls.

- Tiny fragments can hide malicious payloads.
- Filter: `ip frag_offset > 0` shows fragmented packets.

* IPv4 vs IPv6

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Format	192.168.1.1	2001::db8::1
Total addresses	~4 billion	340 undecillion
Header size	20 bytes	40 bytes
Fragmentation	Routers can fragment	Only source fragments

* Protocol Numbers (Memorise!)

Number	Protocol
1	ICMP
6	TCP
17	UDP
47	GRE (tunneling)
50	ESP (IPSec)

* Traceroute — How It Works

- Traceroute uses TTL cleverly to map the network path:
- Step 1: Send packet with $TTL = 1 \rightarrow$ first router drops it, sends ICMP back.
- Step 2: Send packet with $TTL = 2 \rightarrow$ second router drops it, sends ICMP back.
- Step 3: Send packet with $TTL = 3 \rightarrow$ third router drops it, sends ICMP back.
.... continues until destination reached.
- Note: Each ICMP TTL - Exceeded reply reveals one router's IP address!

* Key Wireshark Filters

- `ip` $\rightarrow$ All IP traffic
- `ip.ttl < 10` $\rightarrow$ Low TTL packets — unusual
- `ip.frag.offset > 0` $\rightarrow$ Fragmented packets
- `icmp.type == 11` $\rightarrow$ TTL exceeded messages
- `ip.src == x.x.x.x` $\rightarrow$ Traffic from specific IP
- `ip.dst == x.x.x.x` $\rightarrow$ Traffic to specific IP

* Diagrams:

1) IP Routing — TTL decrements at each hop

Your PC TTL = 64



Home Router TTL = 63



ISP Router TTL = 62



Backbone TTL = 61



Google TTL = 60

2) IP Datagram — key header fields

TTL 8 bits — hops left

Protocol TCP = 6 UDP = 17 ICMP = 1

Source IP
32 bits - origin

Destination IP
32 bits - target

3) How Traceroute works - TTL trick

Send TTL = 1
First router replies

↓
Send TTL = 2
Second router replies

↓
Send TTL = 3
Third router replies

↓
Destination! ~~IP~~
Full path mapped

4) TTL starting values

Windows
TTL = 128

Linux
TTL = 64

CISCO
TTL = 25

* DFIR filters

- $ip.ttl < 10 \rightarrow$ unusual low TTL
- $ip.frag.offset > 0 \rightarrow$ fragmented
- $icmp.type == 11 \rightarrow$ TTL exceeded

* MTU = 1500 bytes (Ethernet) — packets larger than this get fragmented

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000			TCP	55	50339 → 443 [ACK] Seq=1 Ac
2	0.007030300			TCP	66	443 → 50339 [ACK] Seq=1 Ac
3	0.555200000			DNS	79	Standard query 0x0174 HTTP
4	0.555906500			DNS	79	Standard query 0x1a03 A cl
5	0.560343800			DNS	199	Standard query response 0x
6	0.560343800			DNS	153	Standard query response 0x
7	0.563681200			QUIC	1292	Initial, DCID=c6066b396578
8	0.564122700			QUIC	1292	Initial, DCID=c6066b396578
9	0.564961100			QUIC	123	0-RTT, DCID=c6066b3965782e
10	0.572128900			QUIC	82	Initial, SCID=e6066b396578
11	0.573317700			QUIC	1292	Initial, SCID=e6066b396578
12	0.574765800			QUIC	1292	Initial, SCID=e6066b396578
13	0.574765800			QUIC	331	Protected Payload (KP0)
14	0.576424400			QUIC	1292	Handshake, DCID=e6066b3965
15	0.577152300			QUIC	1288	Protected Payload (KP0), DI
16	0.577195900			QUIC	994	Protected Payload (KP0)
17	0.577330700			QUIC	1266	Protected Payload (KP0), DI
18	0.584396200			QUIC	169	Protected Payload (KP0)
19	0.584396200			QUIC	71	Protected Payload (KP0)
20	0.584396200			QUIC	67	Protected Payload (KP0)
21	0.584952900			QUIC	73	Protected Payload (KP0), DI
22	0.624374500			QUIC	64	Protected Payload (KP0)
23	0.624899800			QUIC	75	Protected Payload (KP0), DI
24	0.716199500			QUIC	732	Protected Payload (KP0)
25	0.716199500			QUIC	713	Protected Payload (KP0)

> Frame 1: Packet, 55 bytes on i
> Ethernet II, Src: TpLinkTechn
> Internet Protocol Version 4, I
> Transmission Control Protocol

- > Frame 1: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits)
- > Ethernet II, Src: , Dst: TPLink_
- ▼ Internet Protocol Version 4, Src: Dst:
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 41
 - Identification: 0xb3e9 (46057)
 - > 010. = Flags: 0x2, Don't fragment
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 128
 - Protocol: TCP (6)
 - Header Checksum: 0xb72e [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address:
 - Destination Address:
 - [Stream index: 0]
- > Transmission Control Protocol, Src Port: , Dst Port: Seq: 1, Ac

Command Prompt

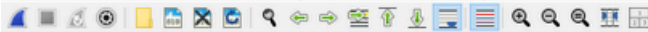
Microsoft Windows [Version 10.0.19045.6466]
 (c) Microsoft Corporation. All rights reserved.

C:\Users\Abhishek>tracert google.com

Tracing route to google.com |
 over a maximum of 30 hops:

1	2 ms	2 ms	2 ms
2	3 ms	3 ms	3 ms
3	5 ms	4 ms	4 ms
4	4 ms	3 ms	10 ms
5	8 ms	6 ms	6 ms
6	5 ms	5 ms	5 ms
7	255 ms	6 ms	5 ms
8	6 ms	5 ms	8 ms

Trace complete.



icmp

No.	Time	Source	Destination	Protocol	Len	ttl	Info
158	26.724398400			ICMP	106		Echo (ping) request id=0x0001, seq=34/8704, ttl=4 (no response received)
159	26.732886800			ICMP	134		Time-to-live exceeded (Time to live exceeded in transit)
160	26.735250500			ICMP	106		Echo (ping) request id=0x0001, seq=35/8960, ttl=4 (no response received)
161	26.738807900			ICMP	134		Time-to-live exceeded (Time to live exceeded in transit)
162	26.739452200			ICMP	106		Echo (ping) request id=0x0001, seq=36/9216, ttl=4 (no response received)
163	26.742792000			ICMP	134		Time-to-live exceeded (Time to live exceeded in transit)
189	32.273027200			ICMP	106		Echo (ping) request id=0x0001, seq=37/9472, ttl=5 (no response received)
190	32.279619600			ICMP	110		Time-to-live exceeded (Time to live exceeded in transit)
191	32.280818300			ICMP	106		Echo (ping) request id=0x0001, seq=38/9728, ttl=5 (no response received)
192	32.286391500			ICMP	110		Time-to-live exceeded (Time to live exceeded in transit)
193	32.286947900			ICMP	106		Echo (ping) request id=0x0001, seq=39/9984, ttl=5 (no response received)
194	32.293192300			ICMP	110		Time-to-live exceeded (Time to live exceeded in transit)
229	37.830907200			ICMP	106		Echo (ping) request id=0x0001, seq=40/10240, ttl=6 (no response received)
230	37.836795200			ICMP	134		Time-to-live exceeded (Time to live exceeded in transit)
231	37.837903700			ICMP	106		Echo (ping) request id=0x0001, seq=41/10496, ttl=6 (no response received)
232	37.842826300			ICMP	134		Time-to-live exceeded (Time to live exceeded in transit)
233	37.844184700			ICMP	106		Echo (ping) request id=0x0001, seq=42/10752, ttl=6 (no response received)
234	37.849529300			ICMP	134		Time-to-live exceeded (Time to live exceeded in transit)
259	43.380798300			ICMP	106		Echo (ping) request id=0x0001, seq=43/11008, ttl=7 (no response received)
260	43.386302700			ICMP	134		Time-to-live exceeded (Time to live exceeded in transit)
261	43.387419000			ICMP	106		Echo (ping) request id=0x0001, seq=44/11264, ttl=7 (no response received)
262	43.394191600			ICMP	134		Time-to-live exceeded (Time to live exceeded in transit)
263	43.395340600			ICMP	106		Echo (ping) request id=0x0001, seq=45/11520, ttl=7 (no response received)
264	43.400752000			ICMP	134		Time-to-live exceeded (Time to live exceeded in transit)